

Supplementary Information

Emotional distress drives change in adolescent problematic digital use and related symptoms

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1. Supplementary Notes

N1. Domain-level strength and temporal directionality share an emotional peak

At the coarse six-domain level, baseline domain strength and longitudinal net directionality both ranked emotional distress first and physical activity last (descriptive Spearman $\rho = 0.71$ across six domains; Table S14). The remaining domain ranks differed, and the item-level minimum-dominating-set analysis identified diet, SDQ and emotional items rather than a domain-level emotional core. The shared emotional peak therefore coexists with a separation between item-level structural coverage and longitudinal prediction.

N2. Exploratory CLPN–DAG directional comparison

The longitudinal CLPN net directionality and the cross-sectional Bayesian DAG net out-degree were positively associated across the six domains (Pearson $r = 0.750$, $P = 0.086$; Spearman $\rho = 0.618$, $P = 0.191$; exact two-sided permutation $P = 0.078$; Table S15). Leave-one-domain-out correlations remained positive ($r = 0.603$ to 0.892 ; Table S16), and DASS lay on the source side in both analyses. The bootstrap confidence interval for Pearson r was wide (-0.604 to 0.995), all DAG skeleton edges were reversible in the CPDAG, and both analyses used SEARCH data. The comparison is therefore exploratory, within-sample and non-causal.

N3. Estimator sensitivity of the structural-control solution

Estimator analyses define the sensitivity of the item-level structural-control solution (Table S17, Figure S20). In the primary cross-validated graphical Lasso network (density = 0.518), exhaustive enumeration identified 10 minimum dominating sets of size three; DIET22, SDQ18 and DASS14 ranked first, second and third by control frequency (0.60 , 0.50 and 0.40 , respectively). Sparser EBICglasso networks (density = 0.246 – 0.284) expanded the MDS to five or six nodes and dispersed driver membership, while the positive-edge network retained DIET22 as a leading driver. These analyses therefore define estimator sensitivity rather than demonstrating invariant driver identity.

N4. Construct-level longitudinal validation in the UK Millennium Cohort Study

Three findings define the construct-level MCS analysis (Table S7, Figure S14). (1) Social-media use was the strongest temporal sink and gaming was also downstream. (2) Self-reported internalising distress was a temporal source, whereas the brief parent-reported SDQ emotional subscale did not show this pattern (net -0.082). (3) Hyperactivity/inattention was the leading source and conduct problems were also upstream. MCS therefore reproduced the downstream position of screen-related behaviour while broadening the source side beyond emotional distress and using non-equivalent instruments.

2. Supplementary Tables

Supplementary Table S1. Analytic samples

Sample	Role	N	Composition	Network nodes
Baseline (Wave 1)	Full cross-sectional network	10,356	52.2% male; mean age 13.96 (SD 2.42)	104 items / 6 domains
Longitudinal matched	Wave 1 ↔ Wave 3 (~1 yr), CLPN	6,092	matched on de-identified hash key	6 domain scores
GMP/MHQ young adults	Cross-sectional construct-level triangulation	147,506	18–24 y; 240 countries	6 mapped domains
MCS (UK) matched	External longitudinal analysis	9,520 (CLPN 7,838)	age 14 → 17; UK Millennium Cohort Study (SN 8156/8682)	6 comparable constructs

Supplementary Table S2. Measurement instruments and item structure (baseline cohort)

Domain	Instrument	Items used	Response scale	Reliability (α)	Role in network
Problematic internet/digital-media use	Chen Internet Addiction Scale, revised (CIAS-R)	26	1–4	0.971	26 nodes
Emotional distress	Depression Anxiety Stress Scales (DASS-21)	21	0–3	0.941	21 nodes
Broad SDQ difficulties	Strengths and Difficulties Questionnaire (SDQ)	25 core items	0–2	0.697	25 nodes
Insomnia	Insomnia Severity Index (ISI)	7	0–4	0.888	7 nodes
Disordered eating	Eating-behaviour questionnaire (DIET)	21 (item 3 excluded)	mixed	0.544	21 nodes
Physical activity	Activity questionnaire (ACT)	4 (items 5–6 excluded)	ordinal	0.622	4 nodes
Total		104			104 nodes

The six SDQ impact-supplement items (items 26–31) were excluded because they assess duration, distress and functional impact of overall difficulties rather than individual symptoms. All 25 core SDQ items were retained; the five prosocial items were reverse-scored so that higher values indicated greater difficulty. The nonparanormal transform was applied to all items before network estimation. Sample Cronbach's alpha values were calculated from baseline item scores (N = 10,356). DIET and ACT showed limited internal consistency, and SDQ was borderline; interpret these domains accordingly.

Supplementary Table S3. Baseline network and control metrics (N = 10,356)

Metric	Value	Interpretation
Nodes / Edges	104 / 2,777	Full six-domain symptom network
Density	0.518	Densely interconnected
Diameter / Mean shortest path	2 / 1.482	Short paths; small-world topology
Clustering / Modularity	0.559 / 0.443	High local clustering; clear module structure
Global efficiency / strength	0.759 / 0.607	Efficient global information transfer
Global average controllability	1.555	Ease of steering towards easy-to-reach states
Global modal controllability	0.929	Ability to steer towards hard-to-reach states
MDS size / equivalent solutions	3 / 10	Three-node driver sets; 10 equivalent solutions
Top driver DIET22 (CF)	0.60	“worried by others about eating too much”
Driver SDQ18 (CF)	0.50	“often lies or cheats”
Driver DASS14 (CF)	0.40	“intolerant of anything keeping me from my task”

All three leading structural drivers lie off the node-strength peak, confirming that structural control is not high centrality.

Supplementary Table S4. Directional profile of the SEARCH cohort (CLPN and Bayesian DAG)

Domain	CLPN net directionality	95% CI	Temporal role	Bayesian DAG net
DASS	+0.249	[0.086, 0.418]	source	+2
ISI	+0.061	[-0.091, 0.217]	ns	+2
CIAS	-0.061	[-0.161, 0.051]	ns	0
DIET	-0.014	[-0.137, 0.108]	ns	-2
SDQ	-0.090	[-0.189, -0.003]	sink	-1
ACT	-0.145	[-0.218, -0.060]	sink	-1

DASS lies on the source side in both the longitudinal CLPN and the exploratory baseline Bayesian graph. Across six domains, the method-level association was Pearson $r = 0.750$ ($P = 0.086$) and was imprecisely estimated.

Supplementary Table S5. Full cross-lagged coefficient matrix, SEARCH cohort (N = 6,092)

W1 ↓ / W3 →	CIAS→	DASS→	SDQ→	DIET→	ISI→	ACT→
CIAS	0.440	0.079	-0.011	0.082	0.081	-0.050
DASS	0.160	0.423	0.120	0.108	0.151	-0.056
SDQ	-0.029	0.050	0.314	0.016	0.004	0.062
DIET	0.073	0.059	0.047	0.218	0.061	0.052
ISI	0.060	0.150	0.070	0.074	0.402	-0.018
ACT	-0.041	-0.010	-0.002	0.027	0.014	0.293

Rows are Wave-1 predictors; columns are Wave-3 outcomes. Diagonal entries are autoregressions; coefficients are standardised (HC3 robust).

Supplementary Table S6. Full cross-lagged coefficient matrix, Millennium Cohort Study (N = 7,838)

Age14 ↓ / Age17 →	Internalising	Conduct	Hyperactivity	Peer	SocialMedia	Gaming
Internalising	0.458	0.019	0.026	0.084	-0.013	-0.045
Conduct	0.014	0.480	0.110	0.074	0.080	0.060
Hyperactivity	-0.001	0.134	0.564	0.052	0.112	0.137
Peer	0.056	0.030	0.059	0.501	-0.063	-0.021
SocialMedia	0.012	0.038	0.007	-0.046	0.213	0.003
Gaming	0	0.018	0.056	0.054	0.050	0.276

Rows are age-14 predictors; columns are age-17 outcomes.

Supplementary Table S7. Millennium Cohort Study net temporal directionality

Construct	Net (out – in)	95% CI	Role
Hyperactivity	+0.177	[0.105, 0.255]	source
Internalising (self-report)	+0.105	[0.023, 0.154]	source
Conduct	+0.098	[0.015, 0.191]	source
Peer	-0.080	[-0.161, -0.004]	sink
Gaming	-0.089	[-0.149, -0.020]	sink
Social-media use	-0.211	[-0.260, -0.139]	strongest sink

Social-media use is the strongest temporal sink and gaming is also downstream. Self-reported internalising distress, hyperactivity/inattention and conduct problems are upstream, with hyperactivity/inattention the leading source.

Supplementary Table S8. Community detection across six algorithms

Algorithm	k	ARI vs domains	NMI vs domains
Louvain	7	0.727	0.843
Leiden	6	0.724	0.837
Walktrap	7	0.615	0.780
Spinglass	7	0.727	0.843
FastGreedy	7	0.727	0.843
Infomap	7	0.666	0.823

Agreement of each algorithm's partition with the six a-priori health domains.

Supplementary Table S9. Most stable directed edges, Bayesian DAG (200 bootstraps)

Directed edge	Bootstrap frequency (200×)
CIAS→ACT	0.970
DASS→SDQ	0.910
CIAS→DIET	0.905
DASS→DIET	0.855
DASS→CIAS	0.850
DASS→ISI	0.625

Directed edge	Bootstrap frequency (200×)
ISI→CIAS	0.525
CIAS→ISI	0.470
CIAS→SDQ	0.440
ISI→DASS	0.375
CIAS→DASS	0.150
SDQ→CIAS	0.125

DASS has several bootstrap-stable outgoing edges. Edge orientations remain exploratory because all skeleton edges are reversible in the CPDAG.

Supplementary Table S10. Principal-component loadings of node-level metrics

Metric	PC1	PC2	PC3
strength	0.182	0.458	0.131
closeness_centrality	0.419	0.024	0.042
betweenness_centrality	0.403	-0.051	0.157
eigenvector_centrality	0.411	0.056	-0.030
average_controllability	0.109	0.476	0.057
modal_controllability	0.102	-0.330	0.302
control_frequency	0.185	-0.093	0.415
participation_coefficient	0.274	-0.013	-0.298
EI1	-0.140	0.445	0.227
EI2	-0.146	0.444	0.216
bridge_strength	0.343	0.120	-0.202
bridge_betweenness	0.403	-0.052	0.158
control_energy_log	-0.085	-0.165	0.660

Control frequency loads on an independent component, separate from traditional centrality and from strength/influence.

Supplementary Table S11. Bayesian DAG net out-degree by model

Domain	Tertile DAG	Continuous Gaussian BN
CIAS	+0	-3
DASS	+2	+3
SDQ	-1	-1
DIET	-2	-4
ISI	+2	+4
ACT	-1	+1

Positive = net source; negative = net sink. DASS is a net source and diet a net sink in both models.

Supplementary Table S12. Baseline comparison, matched longitudinal subsample vs unmatched remainder

Domain	Matched mean (N=6,092)	Unmatched mean (N=4,264)	Standardised mean difference
CIAS	42.21	47.54	-0.351
DASS	8.39	11.94	-0.350
SDQ	18.79	19.73	-0.178
DIET	20.34	21.73	-0.257
ISI	4.32	5.84	-0.308
ACT	10.63	10.18	+0.098

Standardised mean difference is Cohen's d (matched minus unmatched); the retained subsample has modestly lower symptom levels.

Supplementary Table S13. Covariate-adjusted and cluster-robust cross-lagged directionality

domain	net_primary	net_covariate_adj	net_cluster_robust
CIAS	-0.061	-0.079	-0.061
DASS	0.249	0.276	0.249
SDQ	-0.09	-0.096	-0.09
DIET	-0.014	-0.007	-0.014
ISI	0.061	0.049	0.061
ACT	-0.145	-0.142	-0.145

DASS is the rank-1 net out-predictor in the primary, covariate-adjusted and cluster-robust models. Non-unique composite linkage keys accounted for 0.052% of baseline candidate records and were excluded before linkage.

Supplementary Table S14. Domain-level structural strength versus net cross-lagged directionality

Domain	Baseline domain strength	CLPN net directionality	Structural rank	Temporal rank
CIAS	0.965	-0.061	2	4
DASS	1.388	0.249	1	1
SDQ	0.732	-0.09	4	5
DIET	0.512	-0.014	5	3
ISI	0.781	0.061	3	2
ACT	0.412	-0.145	6	6

Aggregated to six domains, structural strength and temporal directionality agree (Spearman rho = 0.71), both ranking emotion first and activity last.

Supplementary Table S15. Exploratory CLPN–DAG directional comparison

statistic	value
Pearson r	0.750
Pearson P (two-sided)	0.086
Spearman rho	0.618
Spearman p	0.191
exact permutation P (two-sided)	0.078
bootstrap CI low	-0.604
bootstrap CI high	0.995

Pearson r = 0.750; Spearman rho = 0.618; exact two-sided permutation P = 0.078.

Supplementary Table S16. Leave-one-domain-out CLPN–DAG correlation

dropped	r_without
CIAS	0.768
DASS	0.603
SDQ	0.726
DIET	0.892
ISI	0.791
ACT	0.731

Leave-one-domain-out correlations remain positive ($r = 0.603$ - 0.892), but the comparison contains only six domains and is not an independent replication.

Supplementary Table S17. Estimator sensitivity of the minimum-dominating-set solution

Network	Density	MDS size	DIET22 CF	DIET22 rank	DASS14 CF	DASS14 rank	SDQ18 CF	SDQ18 rank
EBIC $\gamma=0.0$	0.284	5	0.25	8	0.0	100	0.25	6
EBIC $\gamma=0.25$	0.267	6	0.0	101	0.083	24	0.05	48
EBIC $\gamma=0.5$	0.246	6	0.15	14	0.067	33	0.217	7
Primary CV-glasso	0.518	3	0.600	1	0.400	3	0.500	2
CV positive- edges- only	0.335	5	0.2	2	0.167	4	0.15	7

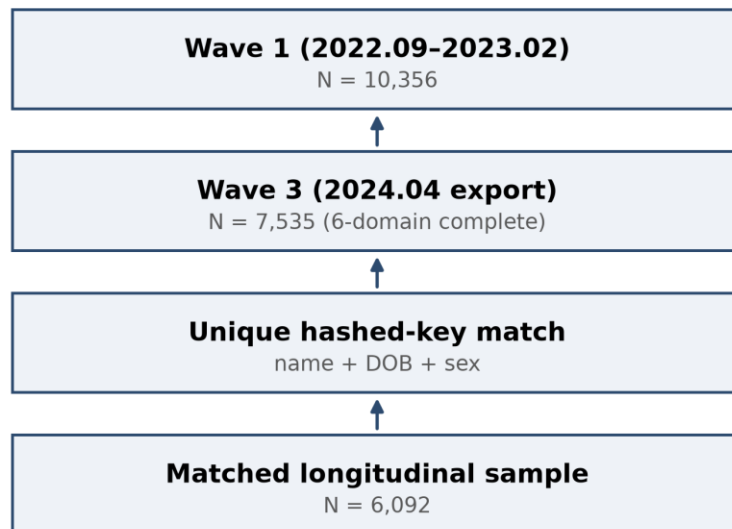
In the primary cross-validated graphical Lasso network (density = 0.518), exhaustive enumeration identified 10 minimum dominating sets of size three. DIET22, SDQ18 and DASS14 ranked first, second and third by control frequency (0.60, 0.50 and 0.40, respectively). For the sensitivity networks, control frequencies were estimated across 60 randomised optimal MDS solutions and therefore quantify estimator sensitivity rather than exact replication of the primary solution.

Supplementary Table S18. Machine-readable files accompanying this supplement

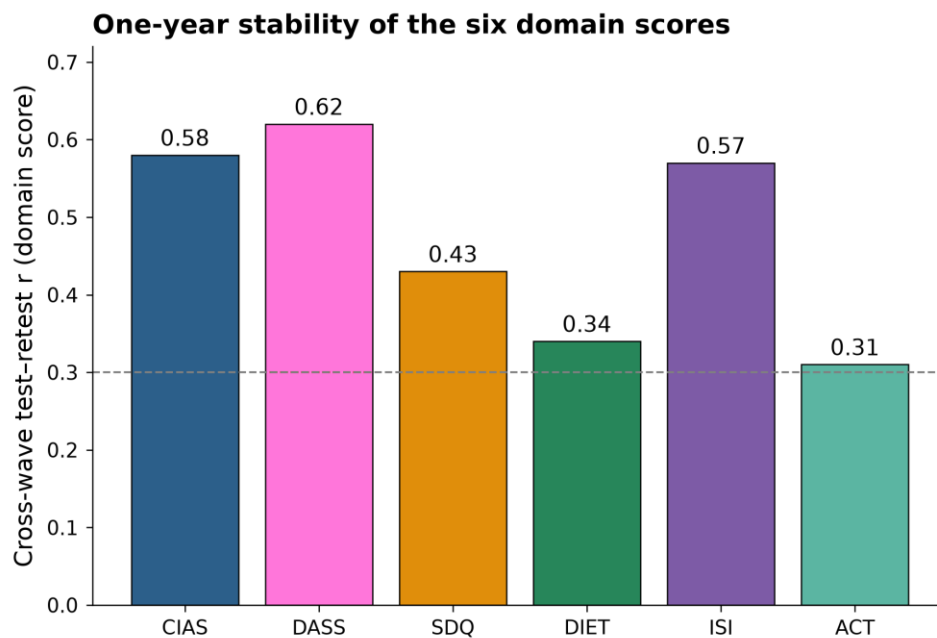
File	Content
clpn_coefficients.csv	Full 6×6 cross-lagged coefficient matrix
clpn_pvalues.csv	Corresponding p-values
clpn_directionality.csv	Net directionality + bootstrap CIs
bn_domain_edges.json	Bayesian DAG edge list
bn_edge_stability.csv	Directed-edge bootstrap frequencies
directed_convergence.csv	CLPN vs. DAG per-domain net directionality
community_multimethod_summary.csv	ARI/NMI for six algorithms
community_evolution_table.csv	Cross-wave community assignments
community_node_stability.csv	Bootstrap co-assignment per node
extended_node_metrics.csv	14 node-level metrics (104 nodes)
method_convergence_corr.csv	Spearman correlations among metrics
method_pca_loadings.csv	PCA loadings (PC1–PC3)
target_controllability.csv	Target-controllability driver sets
driver_stability.csv	300-bootstrap driver selection/CF
lambda_sensitivity.csv	Density/MDS across $\lambda = 0.04$ – 0.10
knockout_efficiency.csv	Efficiency under four attack orderings
dynamics_summary.csv	Steady-state activation by module
mhq_network_pcor.csv	MHQ six-domain partial-correlation matrix
scale_crosswalk.csv	104-node Wave-1↔Wave-3 crosswalk
mcs_clpn_coefficients.csv	MCS 6×6 cross-lagged coefficient matrix
mcs_clpn_directionality.csv	MCS net directionality + bootstrap CIs
clpn_sensitivity.csv	CLPN net directionality under alternative edge/domain definitions
network_estimation_robustness.csv	Estimator-specific density, MDS size and driver-frequency summaries

3. Supplementary Figures

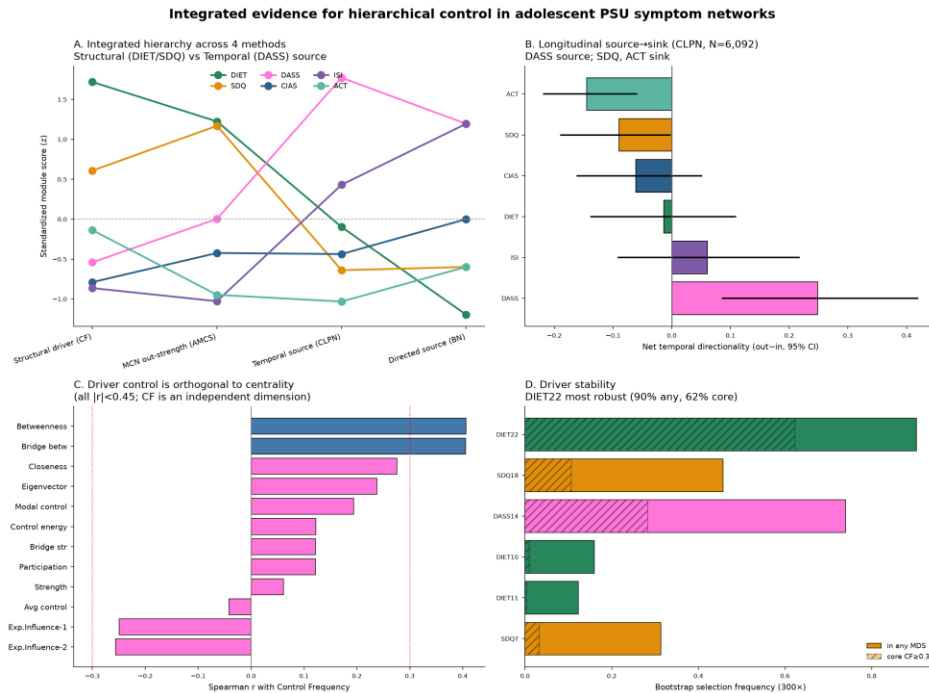
Two-wave individual matching (~1 year apart)



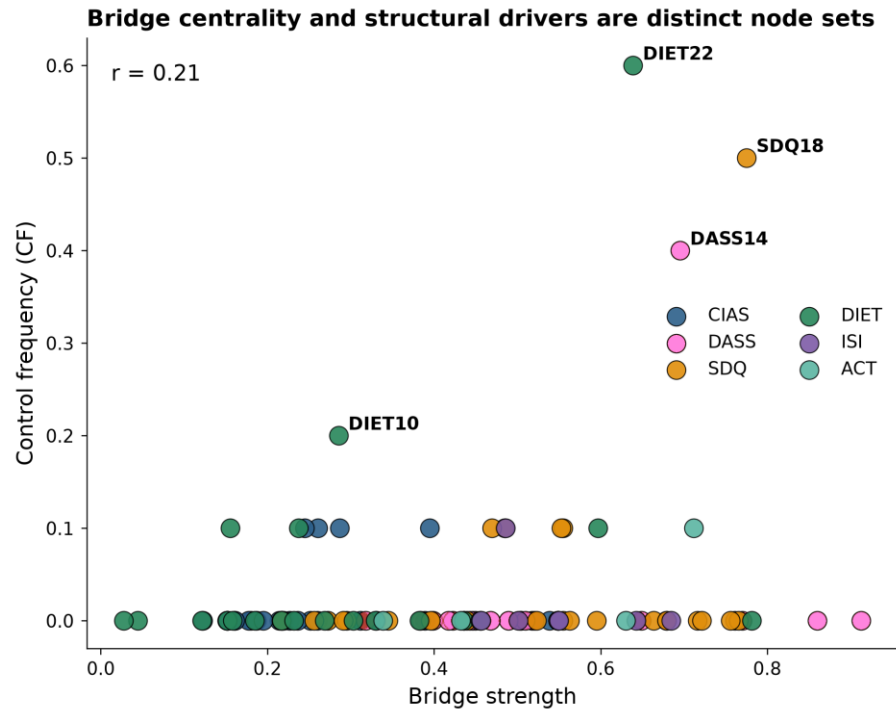
Supplementary Figure S1. Two-wave matching flow. From Wave 1 (N = 10,356) through the Wave-3 six-domain-complete export to the hashed-key-matched longitudinal sample (N = 6,092). DOB, date of birth; arrows indicate the direction of record matching.



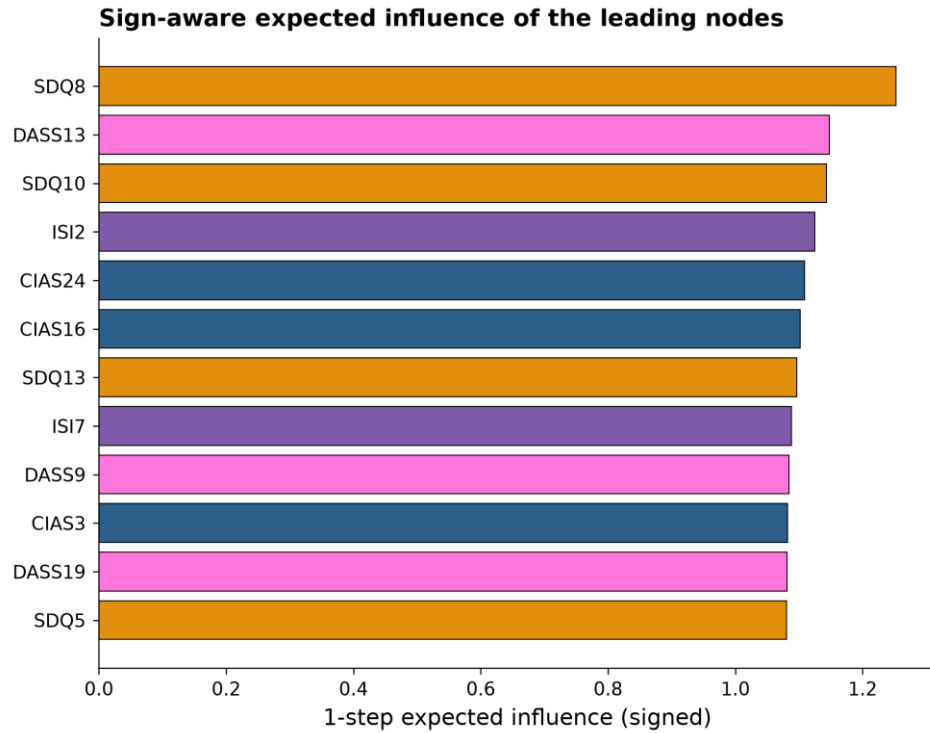
Supplementary Figure S2. One-year domain-score test-retest stability. Cross-wave correlation of each domain score over the one-year interval.



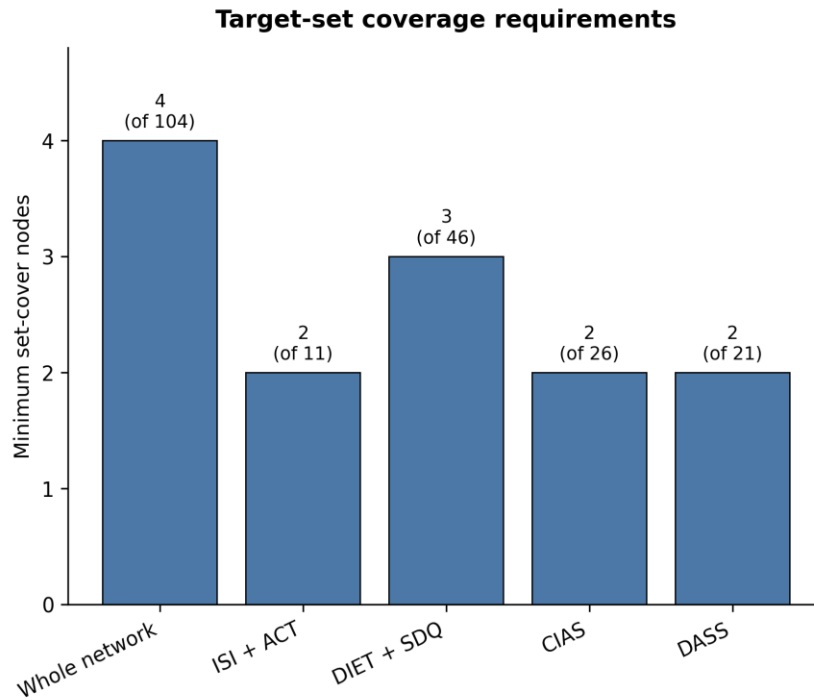
Supplementary Figure S3. Comparison across non-equivalent network metrics. (a) Standardised module scores across four hierarchy definitions (structural-driver CF, module out-strength AMCS, longitudinal CLPN net out-prediction, directed BN source). (b) Longitudinal net directionality with 95% CI. (c) Control frequency versus every traditional centrality metric (all $|r| < 0.45$). (d) Bootstrap driver stability; DIET22 most robust. In the embedded figure title, “PSU” denotes generalised problematic internet and digital-media use assessed with the Chen Internet Addiction Scale–Revised (CIAS-R), rather than smartphone-specific use. CF, control frequency; AMCS, average module control strength; MCN, module control network; CLPN, cross-lagged panel network; BN, Bayesian network; MDS, minimum dominating set; CI, confidence interval. Hatched segments indicate $CF \geq 0.30$, and full bars indicate selection into any MDS. Red dotted lines in c mark correlations of ± 0.30 .



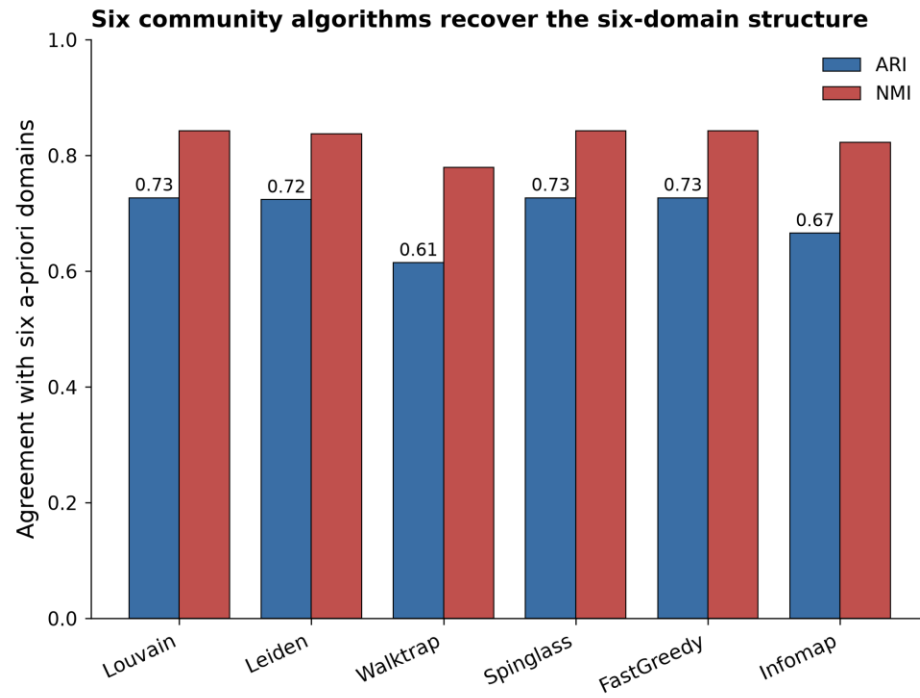
Supplementary Figure S4. Bridge strength versus control frequency: bridge centrality and structural drivers are distinct node sets ($r = 0.21$). CF, control frequency; r , Spearman's correlation coefficient.



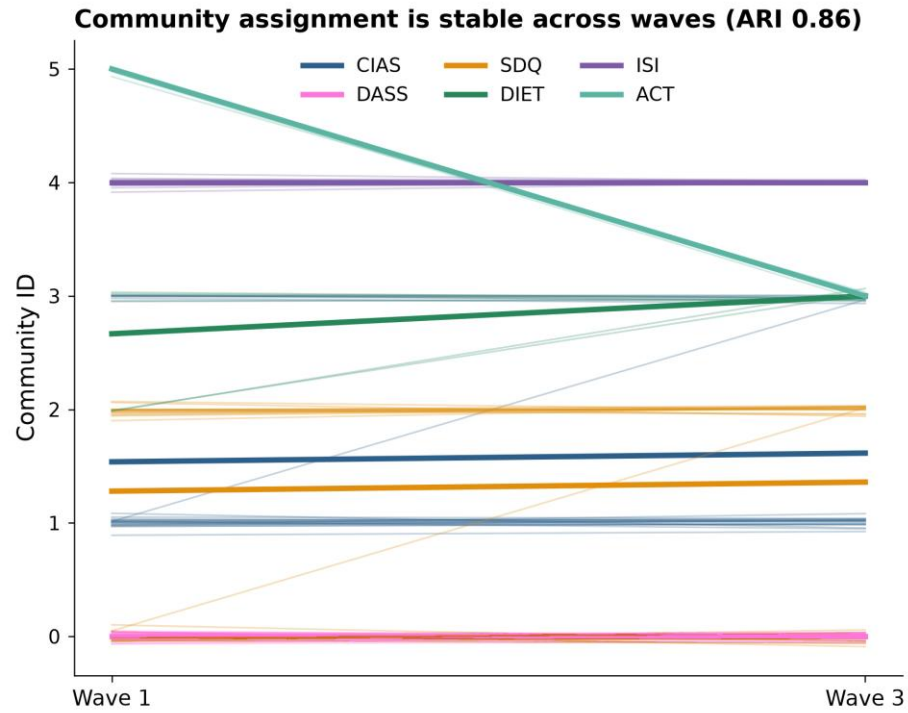
Supplementary Figure S5. One-step signed expected influence of the highest-ranked nodes.



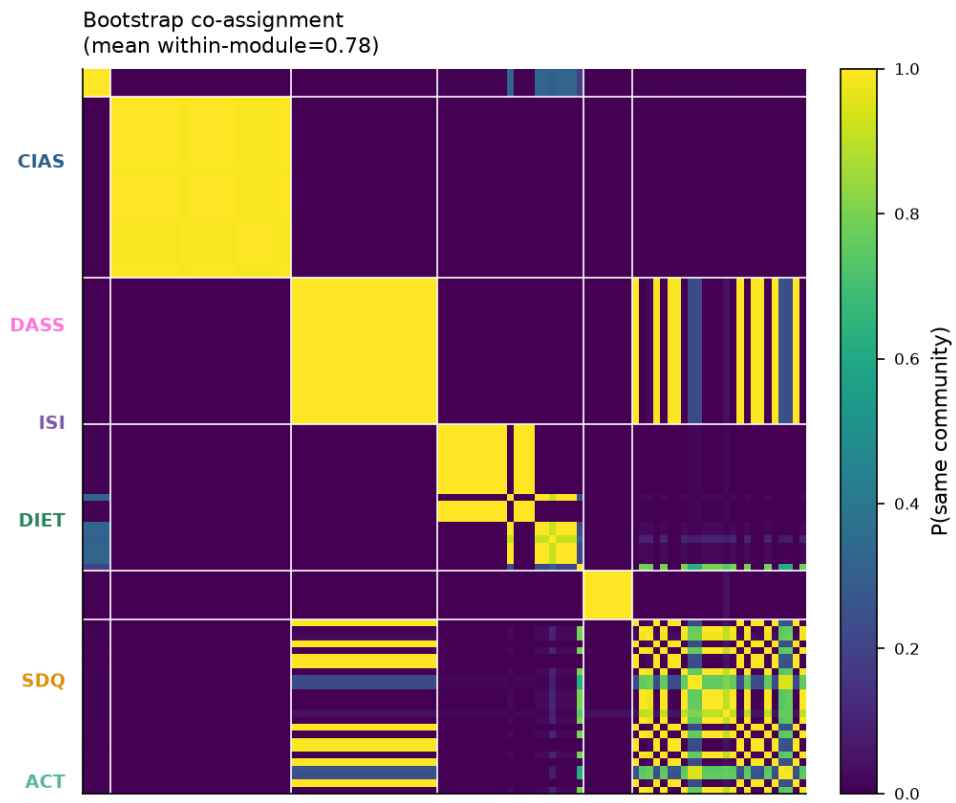
Supplementary Figure S6. Target controllability: minimum drivers needed to control the whole network versus targeted subsets. In the figure, “PSU (CIAS)” denotes generalised problematic internet and digital-media use assessed with the CIAS-R, rather than smartphone-specific use. Values are reported as the minimum number of driver nodes, with the number of target nodes shown in parentheses.



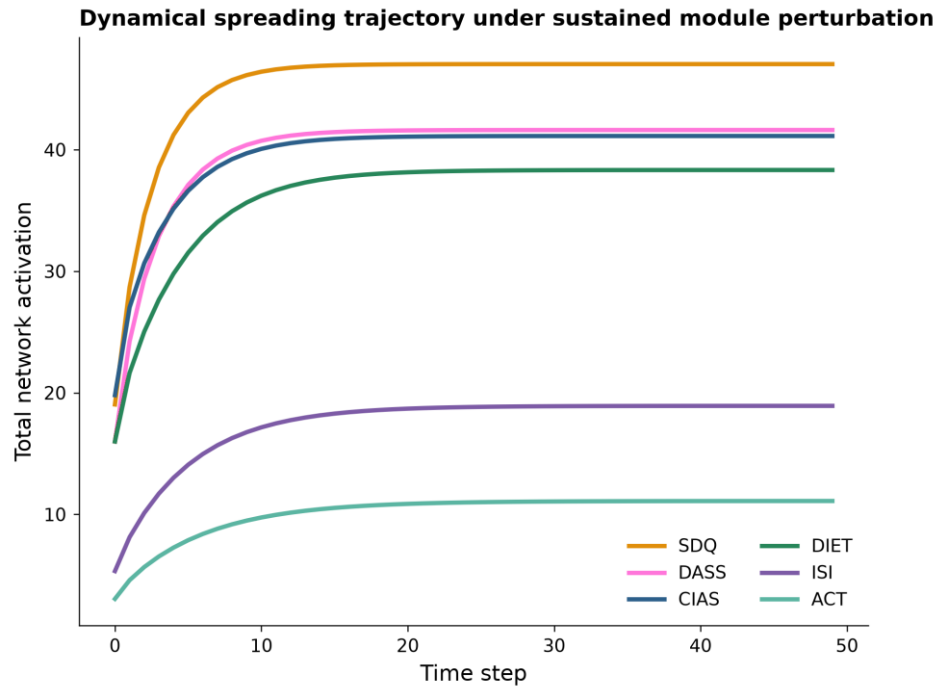
Supplementary Figure S7. Community-algorithm agreement: six community-detection algorithms recover the six a-priori domains (ARI, NMI). ARI, adjusted Rand index; NMI, normalised mutual information.



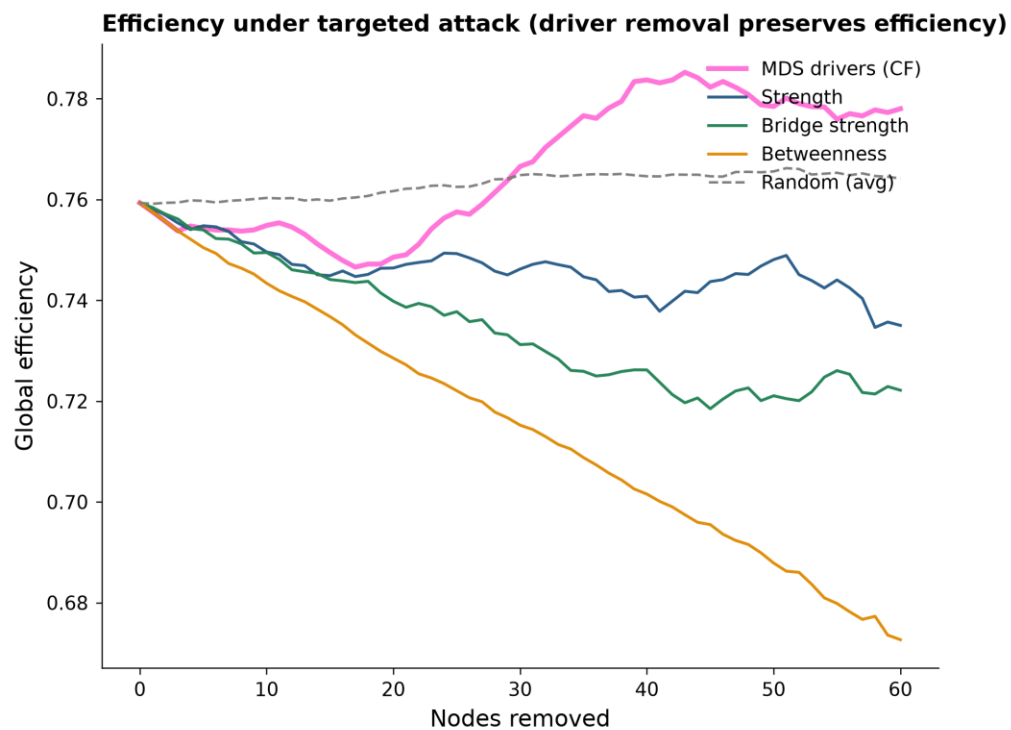
Supplementary Figure S8. Cross-wave community evolution (ARI 0.86; 8 of 89 nodes moved). ARI, adjusted Rand index. Thin lines represent individual nodes and thick lines represent domain-level mean trajectories; community IDs are categorical labels without ordinal meaning.



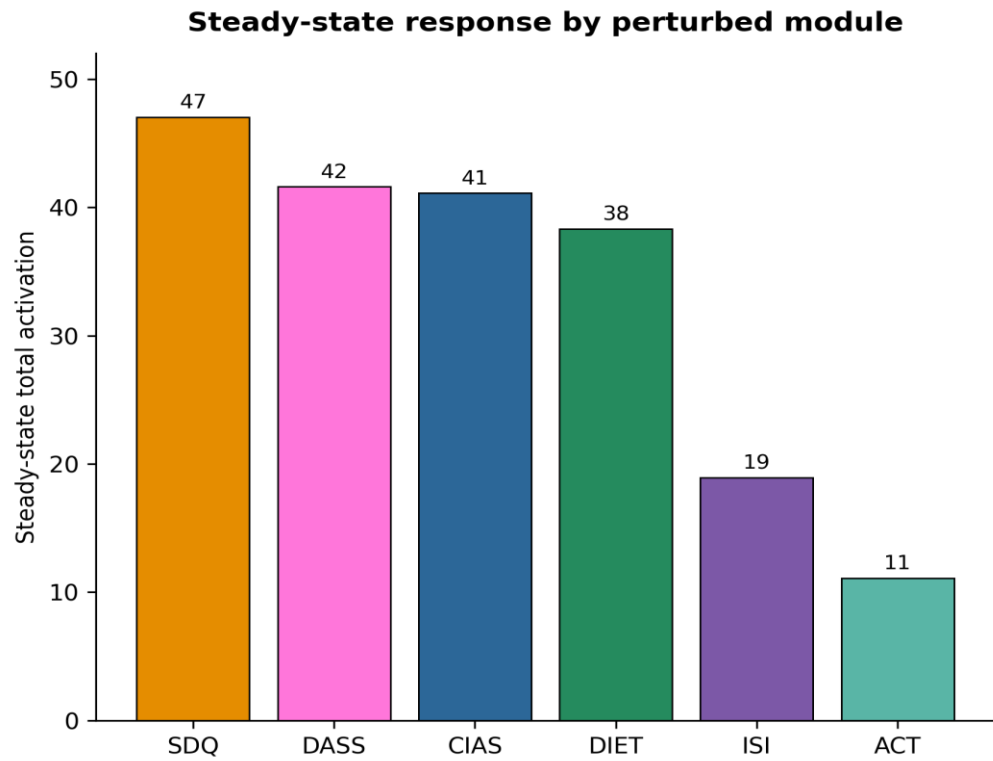
Supplementary Figure S9. Bootstrap co-assignment matrix (mean within-module probability 0.78). Colours indicate bootstrap co-assignment probability from 0 to 1; white boundaries delineate the six prespecified domains. P, probability.



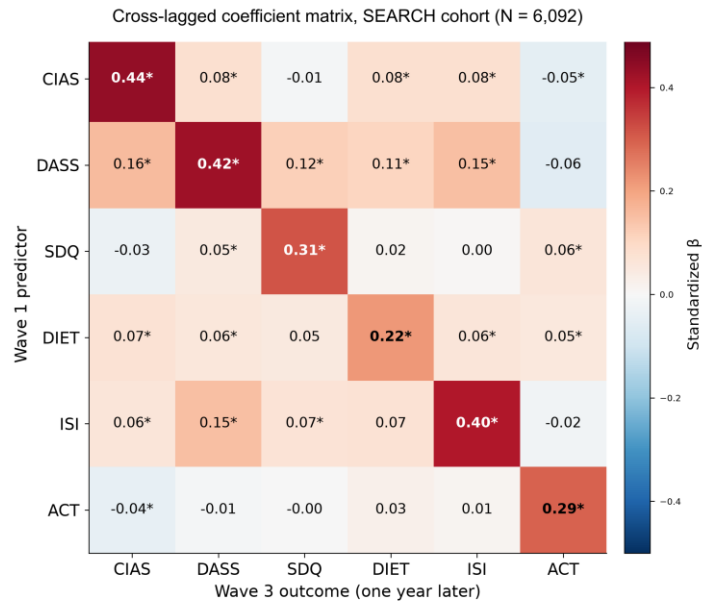
Supplementary Figure S10. Dynamical spreading trajectory: total network activation over time under sustained perturbation of each module.



Supplementary Figure S11. Global efficiency as nodes are removed by different importance metrics.

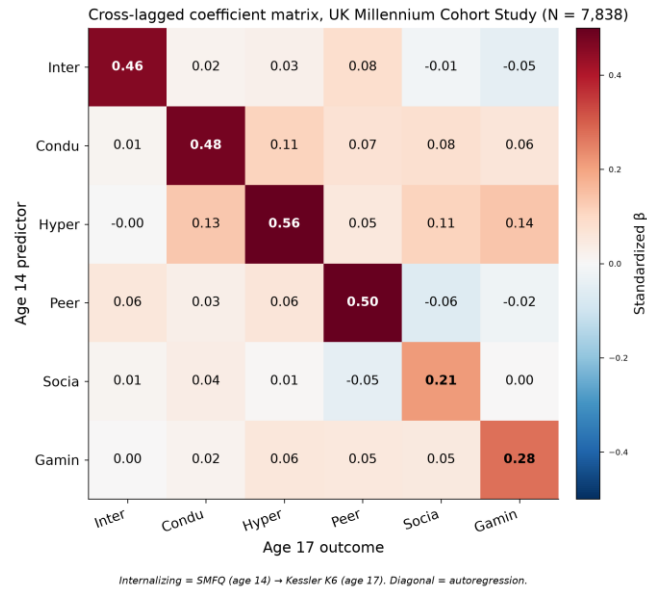


Supplementary Figure S12. Steady-state activation under sustained module perturbation. SDQ and DASS inputs produce greater total activation than ISI and ACT inputs in the specified nonlinear propagation model.

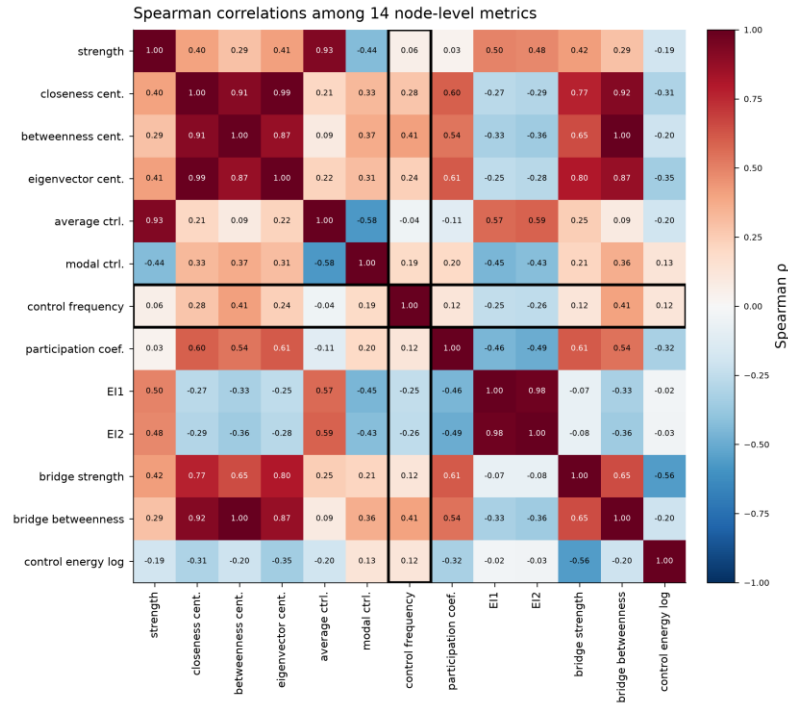


Diagonal = autoregression; * = Bonferroni-significant ($\alpha = 0.0017$). Row DASS shows the strongest off-diagonal out-going effects.

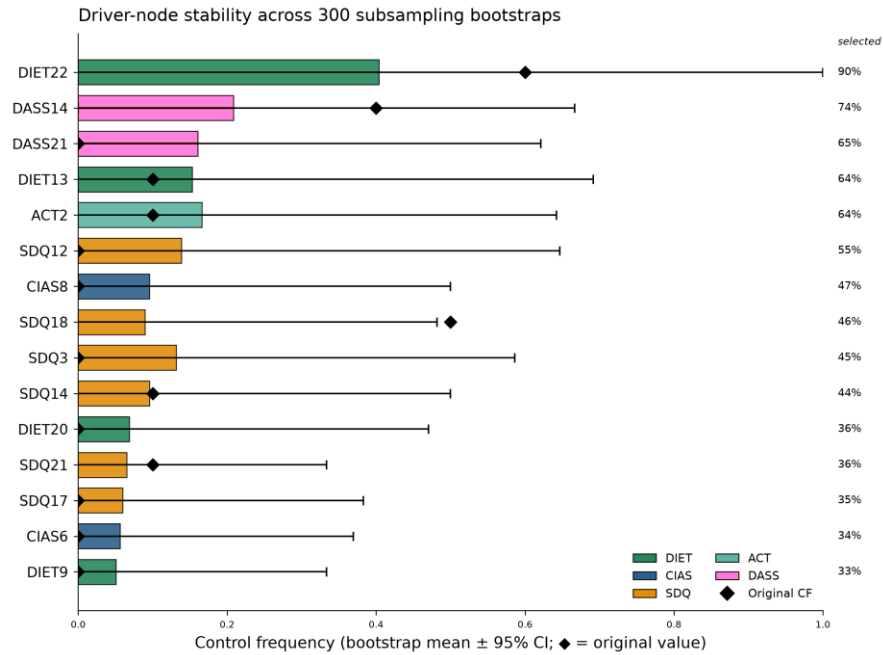
Supplementary Figure S13. Full cross-lagged coefficient matrix, SEARCH cohort (N = 6,092). Rows are Wave-1 predictors, columns are Wave-3 outcomes one year later. Diagonal entries are autoregressions; asterisks mark Bonferroni-significant coefficients ($\alpha = 0.0017$). The DASS row carries the strongest off-diagonal outgoing effects, placing emotion on the source side of the longitudinal network. Warm and blue cells indicate positive and negative coefficients, respectively.



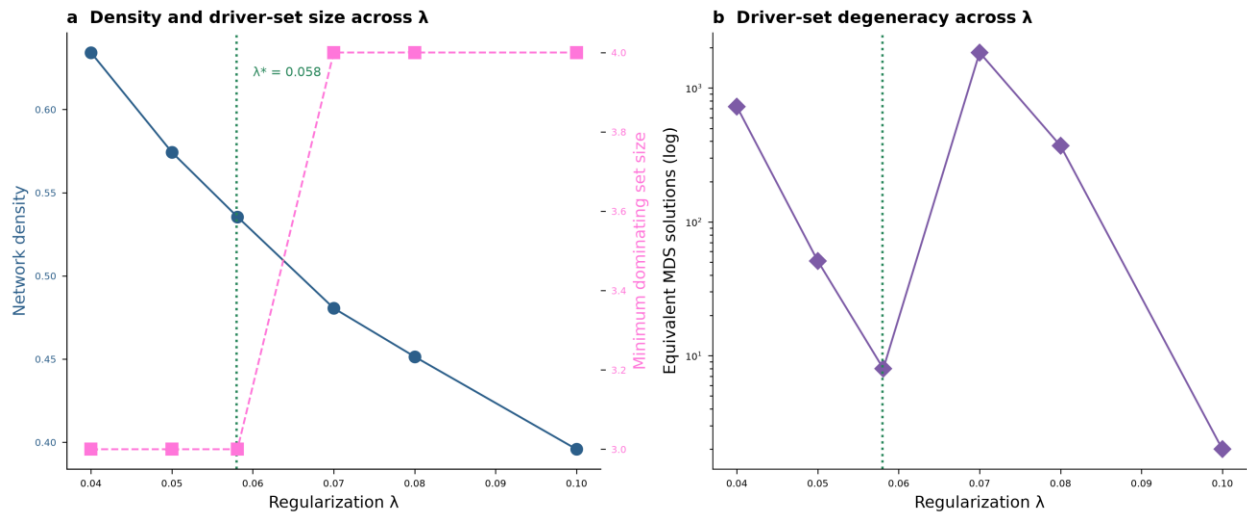
Supplementary Figure S14. Full cross-lagged coefficient matrix, UK Millennium Cohort Study (N = 7,838). SMFQ, Short Mood and Feelings Questionnaire; K6, six-item Kessler Psychological Distress Scale; β , standardised cross-lagged coefficient. Warm and blue cells indicate positive and negative coefficients, respectively. Rows are age-14 predictors, columns are age-17 outcomes. Internalising distress is measured by the SMFQ at age 14 and the Kessler K6 at age 17. The pattern provides construct-level evidence that social-media use and gaming are downstream while broadening the source side to include hyperactivity/inattention and conduct problems.



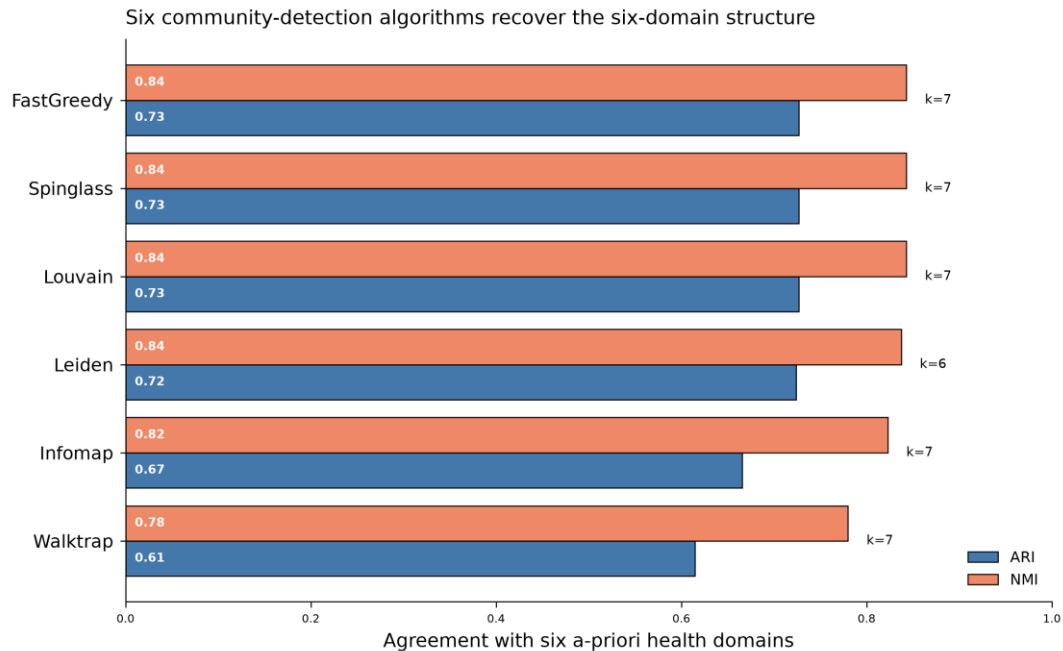
Supplementary Figure S15. Spearman correlations among 14 node-level metrics (104 nodes). The control-frequency row and column (boxed) are weakly correlated with every traditional centrality metric, confirming that structural control is a distinct dimension. Warm and blue cells indicate positive and negative correlations, respectively.



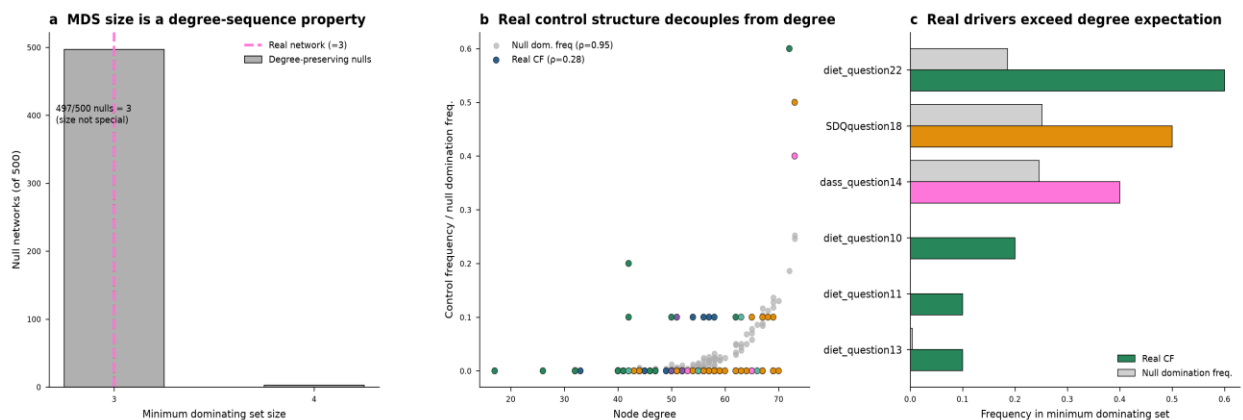
Supplementary Figure S16. Driver-node stability across 300 subsampling bootstraps. Bars show the bootstrap mean control frequency, whiskers the 95% CI, and diamonds the original value; right-hand percentages are the fraction of resamples selecting each node into any minimum dominating set. DIET22 is the most stable driver; SDQ18 is treated as a secondary driver.



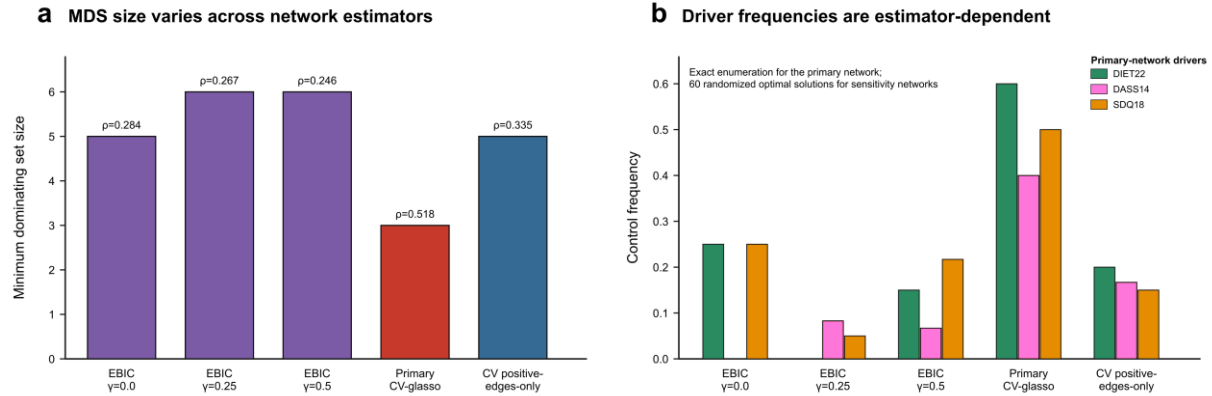
Supplementary Figure S17. Regularisation sensitivity of the network and minimum-dominating-set solution. (a) Network density and minimum-dominating-set size across the regularisation path, with the cross-validated $\lambda^* = 0.058$ marked. (b) Number of equivalent minimum-dominating-set solutions across λ (log scale). Several leading nodes recur between $\lambda = 0.04$ and 0.08 , while set size and membership vary across the wider path.



Supplementary Figure S18. Six community-detection algorithms recover the six-domain structure. ARI and NMI between each algorithm's data-driven partition and the six a-priori domains; k is the number of communities returned. Leiden returns exactly six communities.



Supplementary Figure S19. Minimum-dominating-set size follows the degree sequence while node frequencies differ from degree-matched expectations. (a) Across 500 degree-preserving null networks the minimum-dominating-set size of 3 is a property of the degree sequence (497/500). (b) Null domination frequency is almost fully determined by degree (grey; $p = 0.95$), whereas real control frequency is only weakly related to degree (coloured; $p = 0.28$). (c) The leading drivers enter a minimum dominating set two to three times more often than their degree-matched expectation.



Supplementary Figure S20. Estimator sensitivity of the structural-control solution. (a) The minimum-dominating-set size is three in the primary cross-validated graphical Lasso network (density = 0.518) and increases to five or six in the sparser EBICglasso and positive-edge sensitivity networks. (b) Control frequencies of the three primary-network drivers across estimators. Primary-network values are based on exhaustive enumeration of all 10 minimum dominating sets; sensitivity-network values are estimates from 60 randomised optimal solutions. The variation across estimators indicates that driver identity is not invariant to network estimation.